

INTERNAL ENGINEERING DOCUMENT

# FSTS

## FSTS Website Operating System(TM)

### Improvement Roadmap & Execution Plan

83

Total Improvements

15

Categories

5

Execution Waves

18

P0 Critical

This document converts all 83 approved improvements into a prioritized, wave-based engineering execution plan for the FSTS-WOS(TM) platform. Items are organized by functional area, assigned a priority tier (P0-P3), and placed into execution waves aligned to the Corsair launch milestone and Website #2 readiness goal.

**Architecture Principle:** Every improvement must be evaluated for reuse across many independent client websites. The FSTS-WOS(TM) is not built around Corsair -- it is a multi-tenant platform that must support service businesses, e-commerce stores, training companies, membership organizations, and any future business type without changing core dashboard code.

Generated August 18, 2026

FSTS Client Dashboard -- For FSTS Team Members Only

EXECUTIVE SUMMARY

Priority Distribution & Execution Waves

P0 -- CRITICAL

18

22% of total

P1 -- REQUIRED

41

49% of total

P2 -- IMPORTANT

18

22% of total

P3 -- ENHANCEMENT

6

7% of total

Wave 1 -- Corsair Critical

Security, payments, authentication, email, live CMS, class/event correctness

? Milestone: Corsair First-Site Production Completion

36 items ? 18 P0

Wave 2 -- Website #2 Readiness

Multi-tenant safeguards, onboarding reliability, reusable seeding, permissions, CI/E2E

? Milestone: Website #2 Ready

32 items

Wave 3 -- Client Self-Service

Reduce FSTS manual support requirements

8 items

Wave 4 -- UX & Performance

Loading states, navigation speed, accessibility, dashboard polish

7 items

Wave 5 -- Growth & Enhancement

Marketing, commerce, advanced automation, and non-blocking improvements

0 items

Table of Contents

|    |                                  |   |    |                              |    |
|----|----------------------------------|---|----|------------------------------|----|
| 01 | User Management & Access Control | 8 | 02 | Permissions & Route Security | 9  |
| 03 | Media & Images                   | 8 | 04 | Email & Notifications        | 11 |
| 05 | Onboarding & Site Setup          | 8 | 06 | Products & Commerce          | 8  |
| 07 | Payments & Checkout              | 3 | 08 | Courses & Classes            | 7  |

|    |                         |   |    |                               |   |
|----|-------------------------|---|----|-------------------------------|---|
| 09 | Services                | 3 | 10 | Flyers & Marketing            | 2 |
| 11 | Sidebar & Navigation    | 1 | 12 | Performance & Build Quality   | 4 |
| 13 | CMS & Live Data         | 2 | 14 | Security & Seeding Safeguards | 3 |
| 15 | Infrastructure & DevOps | 6 |    |                               |   |

## WAVE 1

**Corsair Critical**

Security, payments, authentication, email, live CMS, class/event correctness

? Milestone: Corsair First-Site Production Completion

| ID   | PRIORITY | CATEGORY                     | IMPROVEMENT                                                                                              | TAGS                         |
|------|----------|------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------|
| #14  | P0       | Media & Images               | Let clients actually use their uploaded images across their whole site                                   | Client UX Revenue            |
| #21  | P0       | Email & Notifications        | Make failed welcome email deliveries visible so admins know when a new registrant didn't get their email | Client UX Reliability        |
| #65  | P0       | Email & Notifications        | Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard         | Client UX Automation         |
| #66  | P0       | Email & Notifications        | Confirm a real email reaches the right inbox once Resend is configured for the first client              | Reliability Client UX        |
| #85  | P0       | Permissions & Route Security | Make the guard check smart enough to catch null checks that don't show an access-denied message          | Security                     |
| #86  | P0       | Permissions & Route Security | Extend the access-denied guard requirement to any new route category added outside the sites folder      | Security Infrastructure      |
| #92  | P0       | Services                     | Confirm services created in the dashboard actually appear on the public website                          | Client UX Reliability        |
| #135 | P0       | CMS & Live Data              | Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds           | Client UX Reliability        |
| #159 | P0       | Permissions & Route Security | Confirm write-path guards work for CRM, forms, and media modules too                                     | Security                     |
| #164 | P0       | Permissions & Route Security | Confirm a client-role user is automatically blocked from every design-locked route                       | Security                     |
| #171 | P0       | Permissions & Route Security | Confirm the Footer, Email, CRM, and Payments pages can't be unlocked by a client user                    | Security                     |
| #172 | P0       | Permissions & Route Security | Catch a client user bypassing the navigation update and delete guards too                                | Security                     |
| #193 | P0       | Courses & Classes            | Make sure a client with a stale session can't register for a class that's already been cancelled         | Revenue Reliability          |
| #202 | P0       | Email & Notifications        | Alert admins the moment a paid order confirmation email permanently fails to deliver                     | Revenue Reliability          |
| #203 | P0       | Payments & Checkout          | Confirm Square payments still process correctly end-to-end after the idempotency changes                 | Revenue Reliability          |
| #204 | P0       | Payments & Checkout          | Harden the provider-agnostic payment webhook with the same idempotency and email-state guards            | Revenue Security Reliability |
| #206 | P0       | Payments & Checkout          | Block checkout when a course or event price is missing or corrupt so no broken order reaches Square      | Revenue Reliability          |

| ID   | PRIORITY | CATEGORY                         | IMPROVEMENT                                                                            | TAGS                       |
|------|----------|----------------------------------|----------------------------------------------------------------------------------------|----------------------------|
| #211 | P0       | Infrastructure & DevOps          | Confirm clients can actually sign in once the Cloudflare DNS fix is applied            | Client UX Infrastructure   |
| #15  | P1       | Media & Images                   | Prevent broken images when a media asset is deleted that's still in use                | Client UX Reliability      |
| #20  | P1       | Infrastructure & DevOps          | Make sure Vercel deploys cleanly against the rewritten commit history                  | Infrastructure             |
| #22  | P1       | Email & Notifications            | Warn admins when the lead alert recipient address looks wrong or bounces               | Reliability Client UX      |
| #23  | P1       | Email & Notifications            | Let admins send a test lead alert to confirm their notification email is working       | Client UX Automation       |
| #26  | P1       | Email & Notifications            | Fix broken tests in the email suite that are silently hiding regressions               | Technical debt Reliability |
| #27  | P1       | Email & Notifications            | Confirm the notification email override reaches the right address end-to-end           | Reliability                |
| #61  | P1       | Infrastructure & DevOps          | Confirm www.fstscientsystem.com actually redirects once the Vercel alias is added      | Infrastructure Client UX   |
| #81  | P1       | Media & Images                   | Auto-generate alt text when images are uploaded so clients never have empty alt text   | Client UX Automation       |
| #82  | P1       | Media & Images                   | Confirm image picker flows work end-to-end in a real browser across every module       | Reliability                |
| #136 | P1       | CMS & Live Data                  | Extend live CMS data to the course detail pages so per-course pricing is also accurate | Client UX Reliability      |
| #137 | P1       | Services                         | Seed services data for Corsair so the live services section appears on the website     | Client UX                  |
| #141 | P1       | Email & Notifications            | Confirm the welcome email fires when a new user is created -- and only then            | Reliability                |
| #143 | P1       | Email & Notifications            | Make sure welcome emails link to the correct dashboard URL after deployment            | Reliability                |
| #157 | P1       | Courses & Classes                | Confirm courses stay hidden when the agency switches the module off platform-wide      | Security Reliability       |
| #181 | P1       | User Management & Access Control | Catch a stale legacy role grant the moment a permission is renamed or removed          | Security                   |
| #189 | P1       | Courses & Classes                | Confirm lifecycle alerts in the edit dialog always reflect live registration data      | Client UX Reliability      |
| #194 | P1       | Courses & Classes                | Catch a flyer that stays published after its linked class is deleted outright          | Client UX Reliability      |
| #195 | P1       | Courses & Classes                | Confirm the lifecycle clock is registered so tick can never be silently skipped        | Reliability Automation     |

## WAVE 2

# Website #2 Readiness

Multi-tenant safeguards, onboarding reliability, reusable seeding, permissions, CI/E2E

? Milestone: Website #2 Ready

| ID   | PRIORITY | CATEGORY                         | IMPROVEMENT                                                                                            | TAGS                       |
|------|----------|----------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| #11  | P1       | User Management & Access Control | Let admins manually unlock a locked portal user account                                                | Client UX Security         |
| #12  | P1       | User Management & Access Control | Stop other user management pages from leaking the full site list                                       | Security Reliability       |
| #62  | P1       | Infrastructure & DevOps          | Turn on the branch protection rule so the identity check can't be bypassed on GitHub                   | Infrastructure Security    |
| #72  | P1       | Onboarding & Site Setup          | Add an onboarding wizard so admins can seed a new client without touching the CLI                      | Automation Client UX       |
| #73  | P1       | Onboarding & Site Setup          | Confirm the seeder runs end-to-end for a second site without corrupting the first                      | Reliability Security       |
| #90  | P1       | Onboarding & Site Setup          | Make sure a second launch on the same session can't silently create a duplicate site                   | Reliability                |
| #91  | P1       | Onboarding & Site Setup          | Confirm the temp domain is assigned to the site record when the client picks a temporary URL           | Client UX Reliability      |
| #94  | P1       | Services                         | Prevent two services from sharing the same slug on the same site                                       | Reliability                |
| #108 | P1       | Products & Commerce              | Make sure filtering products by category can't silently return wrong results                           | Reliability                |
| #109 | P1       | Products & Commerce              | Confirm disabled-module sites can't have their products scraped via the public endpoint                | Security                   |
| #112 | P1       | Products & Commerce              | Catch a future refactor that breaks the products list endpoint's visibility filter                     | Reliability Technical debt |
| #118 | P1       | Products & Commerce              | Confirm featured products can't appear in the featured spotlight if they're still hidden               | Client UX Reliability      |
| #123 | P1       | Products & Commerce              | Prevent two products from sharing the same slug on the same site                                       | Reliability                |
| #129 | P1       | Security & Seeding Safeguards    | Confirm the seed gate can't be bypassed if SEED_ALLOWED is set to 'false' or empty                     | Security                   |
| #130 | P1       | Security & Seeding Safeguards    | Make sure the add-on catalog seed can't run on production even if someone sets SEED_ALLOWED by mistake | Security                   |
| #131 | P1       | Security & Seeding Safeguards    | Confirm visitors can't forge form submissions for a site they don't belong to                          | Security                   |
| #144 | P1       | User Management & Access Control | Confirm addSiteRole stays blocked even if superadmin status is set after roles already exist           | Security                   |

| ID   | PRIORITY | CATEGORY                         | IMPROVEMENT                                                                                      | TAGS                          |
|------|----------|----------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------|
| #145 | P1       | User Management & Access Control | Catch a superadmin partial update silently wiping roles on any site owner                        | Security Reliability          |
| #146 | P1       | User Management & Access Control | Confirm role changes made through users.update are always recorded in the activity log           | Security Reliability          |
| #158 | P1       | Courses & Classes                | Confirm a disabled-module site can't be unlocked by a direct courses.get call                    | Security                      |
| #184 | P1       | Permissions & Route Security     | Confirm the module-access grid for site users also covers every role                             | Security                      |
| #185 | P1       | Permissions & Route Security     | Confirm the module-access matrix stays complete when a dashboard module is added or removed      | Security Infrastructure       |
| #186 | P1       | Permissions & Route Security     | Confirm the permissions preview always matches the role actually being saved                     | Client UX Security            |
| #13  | P2       | User Management & Access Control | Confirm portal lockout and password flow can't be broken by a future change                      | Reliability Technical debt    |
| #16  | P2       | Media & Images                   | Clean up legacy base64 images already in clients' media libraries                                | Technical debt Infrastructure |
| #30  | P2       | Infrastructure & DevOps          | Make the client-website E2E suite actually run in CI against a deployed site                     | Infrastructure Reliability    |
| #44  | P2       | Media & Images                   | Confirm the purge tool is safe to run when a site has thousands of records across multiple pages | Reliability                   |
| #58  | P2       | Infrastructure & DevOps          | Make sure the workspace branch can't silently drift ahead of origin/main again                   | Infrastructure Technical debt |
| #84  | P2       | Performance & Build Quality      | Confirm a dropped skeleton guard shows up as a failing test -- not a silent blank page           | Technical debt Reliability    |
| #93  | P2       | Onboarding & Site Setup          | Seed placeholder services when onboarding creates a site with services enabled                   | Client UX                     |
| #114 | P2       | Products & Commerce              | Confirm seeded product titles always carry the client's business name                            | Client UX                     |
| #192 | P2       | Onboarding & Site Setup          | Add a CONTRIBUTING guide so new contributors always commit with the right identity               | Infrastructure Technical debt |

WAVE 3

## Client Self-Service

Reduce FSTS manual support requirements

| ID   | PRIORITY | CATEGORY                         | IMPROVEMENT                                                                                      | TAGS                 |
|------|----------|----------------------------------|--------------------------------------------------------------------------------------------------|----------------------|
| #45  | P2       | Media & Images                   | Let clients fix broken images directly from the nav badge -- not just see the count              | Client UX Automation |
| #107 | P2       | Products & Commerce              | Let visitors see which product categories exist before filtering                                 | Client UX Revenue    |
| #142 | P2       | Email & Notifications            | Let admins re-send the welcome email to a pending user who hasn't signed in yet                  | Client UX Automation |
| #167 | P2       | Flyers & Marketing               | Let clients duplicate an existing flyer instead of rebuilding from scratch                       | Client UX Automation |
| #168 | P2       | Flyers & Marketing               | Confirm health notifications give a clear 'access denied' message to non-admin users             | Client UX Security   |
| #169 | P2       | Products & Commerce              | Make sure client admins can manage their Square discounts without needing FSTS staff help        | Client UX Revenue    |
| #187 | P2       | User Management & Access Control | Let site owners invite dashboard users directly from their site -- without needing a super admin | Client UX Automation |
| #207 | P2       | Courses & Classes                | Let admins fix a broken course link on an event from the warning banner -- not just see it       | Client UX Automation |

## WAVE 4

## UX & Performance

Loading states, navigation speed, accessibility, dashboard polish

| ID   | PRIORITY | CATEGORY                    | IMPROVEMENT                                                                                   | TAGS                             |
|------|----------|-----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------|
| #83  | P2       | Performance & Build Quality | Extend loading-skeleton guards to the remaining dashboard pages not yet covered               | Client UX                        |
| #54  | P3       | Performance & Build Quality | Fix the circular vendor chunk warning so every production build is clean                      | Technical debt<br>Infrastructure |
| #55  | P3       | Performance & Build Quality | Make page transitions feel instant by preloading chunks on hover                              | Client UX                        |
| #80  | P3       | Media & Images              | Verify the image editor is keyboard-navigable and screen-reader friendly                      | Client UX                        |
| #110 | P3       | Onboarding & Site Setup     | Remember that the getting-started nudge was dismissed so it doesn't reappear on every visit   | Client UX                        |
| #111 | P3       | Onboarding & Site Setup     | Confirm the placeholder nudge banner shows and disappears at the right times                  | Client UX                        |
| #176 | P3       | Sidebar & Navigation        | Add a Role Permissions link to the per-site sidebar so admins can reach it from inside a site | Client UX                        |

WAVE 5

# Growth & Enhancement

Marketing, commerce, advanced automation, and non-blocking improvements

| ID | PRIORITY | CATEGORY | IMPROVEMENT | TAGS |
|----|----------|----------|-------------|------|
|----|----------|----------|-------------|------|

8 improvements

## User Management & Access Control

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                     | WHY IT MAKES THE DASHBOARD BETTER                                                            | TAGS                       |
|---|------|----------|------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------|
| 1 | #11  | P1       | W2   | Let admins manually unlock a locked portal user account                                          | Admins can unblock users without developer intervention, reducing support delays.            | Client UX Security         |
| 2 | #12  | P1       | W2   | Stop other user management pages from leaking the full site list                                 | Prevents unauthorized data exposure by scoping site lists to the correct context.            | Security Reliability       |
| 3 | #13  | P2       | W2   | Confirm portal lockout and password flow can't be broken by a future change                      | Guards against regressions that could lock users out permanently.                            | Reliability Technical debt |
| 4 | #144 | P1       | W2   | Confirm addSiteRole stays blocked even if superadmin status is set after roles already exist     | Closes a privilege escalation window where late-granted superadmin could bypass role guards. | Security                   |
| 5 | #145 | P1       | W2   | Catch a superadmin partial update silently wiping roles on any site owner                        | Prevents accidental data loss that could lock site owners out of their own accounts.         | Security Reliability       |
| 6 | #146 | P1       | W2   | Confirm role changes made through users.update are always recorded in the activity log           | Full audit trail for every permission change so admins can trace who changed what and when.  | Security Reliability       |
| 7 | #181 | P1       | W1   | Catch a stale legacy role grant the moment a permission is renamed or removed                    | Prevents ghost permissions that outlive their intended purpose from accumulating over time.  | Security                   |
| 8 | #187 | P2       | W3   | Let site owners invite dashboard users directly from their site -- without needing a super admin | Eliminates the bottleneck of requiring FSTS staff involvement for every user invitation.     | Client UX Automation       |

9 improvements

## Permissions & Route Security

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                        | WHY IT MAKES THE DASHBOARD BETTER                                                             | TAGS                       |
|---|------|----------|------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------|
| 1 | #85  | P0       | W1   | Make the guard check smart enough to catch null checks that don't show an access-denied message     | Ensures users always see a clear 'access denied' message instead of a confusing blank screen. | Security                   |
| 2 | #86  | P0       | W1   | Extend the access-denied guard requirement to any new route category added outside the sites folder | Future-proofs the app so new routes automatically inherit the correct access-denied behavior. | Security<br>Infrastructure |
| 3 | #159 | P0       | W1   | Confirm write-path guards work for CRM, forms, and media modules too                                | Closes gaps where CRM or media write operations could bypass module-access checks.            | Security                   |
| 4 | #164 | P0       | W1   | Confirm a client-role user is automatically blocked from every design-locked route                  | Enforces the client permission boundary so clients can't reach admin-only design tools.       | Security                   |
| 5 | #171 | P0       | W1   | Confirm the Footer, Email, CRM, and Payments pages can't be unlocked by a client user               | Hardens the client boundary on critical business and payment configuration pages.             | Security                   |
| 6 | #172 | P0       | W1   | Catch a client user bypassing the navigation update and delete guards too                           | Ensures mutation guards on nav items cover all roles, not just the read-path checks.          | Security                   |
| 7 | #184 | P1       | W2   | Confirm the module-access grid for site users also covers every role                                | Guarantees no role is accidentally excluded from the module-access matrix.                    | Security                   |
| 8 | #185 | P1       | W2   | Confirm the module-access matrix stays complete when a dashboard module is added or removed         | Prevents new modules from launching without any permission boundaries set.                    | Security<br>Infrastructure |
| 9 | #186 | P1       | W2   | Confirm the permissions preview always matches the role actually being saved                        | Admins see exactly what they're granting before they save -- no surprise permission gaps.     | Client UX<br>Security      |

8 improvements

## Media & Images

| # | ID  | PRIORITY | WAVE | WHAT IT DOES                                                                                     | WHY IT MAKES THE DASHBOARD BETTER                                                                          | TAGS                          |
|---|-----|----------|------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1 | #14 | P0       | W1   | Let clients actually use their uploaded images across their whole site                           | Unlocks the media library as a shared asset pool, so clients get full value from every upload.             | Client UX Revenue             |
| 2 | #15 | P1       | W1   | Prevent broken images when a media asset is deleted that's still in use                          | Eliminates embarrassing broken-image placeholders on client-facing websites.                               | Client UX Reliability         |
| 3 | #16 | P2       | W2   | Clean up legacy base64 images already in clients' media libraries                                | Reduces database size and speeds up media library load times across all client sites.                      | Technical debt Infrastructure |
| 4 | #44 | P2       | W2   | Confirm the purge tool is safe to run when a site has thousands of records across multiple pages | Prevents partial deletes that would leave orphaned records when running cleanup at scale.                  | Reliability                   |
| 5 | #45 | P2       | W3   | Let clients fix broken images directly from the nav badge -- not just see the count              | Turns the broken-image badge into an actionable fix flow instead of a passive warning.                     | Client UX Automation          |
| 6 | #80 | P3       | W4   | Verify the image editor is keyboard-navigable and screen-reader friendly                         | Makes the image editor accessible to users who rely on keyboards or assistive technology.                  | Client UX                     |
| 7 | #81 | P1       | W1   | Auto-generate alt text when images are uploaded so clients never have empty alt text             | Improves SEO and accessibility on every client website automatically, with zero extra effort from clients. | Client UX Automation          |
| 8 | #82 | P1       | W1   | Confirm image picker flows work end-to-end in a real browser across every module                 | Catches browser-specific picker bugs before clients encounter them on live sites.                          | Reliability                   |

11 improvements

## Email & Notifications

| #  | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                             | WHY IT MAKES THE DASHBOARD BETTER                                                               | TAGS                          |
|----|------|----------|------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------|
| 1  | #21  | P0       | W1   | Make failed welcome email deliveries visible so admins know when a new registrant didn't get their email | Admins can proactively follow up instead of discovering the issue after a registrant complains. | Client UX<br>Reliability      |
| 2  | #22  | P1       | W1   | Warn admins when the lead alert recipient address looks wrong or bounces                                 | Catches misconfigured alert emails before real leads go unnoticed.                              | Reliability<br>Client UX      |
| 3  | #23  | P1       | W1   | Let admins send a test lead alert to confirm their notification email is working                         | One-click confidence check so admins know leads are flowing to the right inbox.                 | Client UX<br>Automation       |
| 4  | #26  | P1       | W1   | Fix broken tests in the email suite that are silently hiding regressions                                 | Restores reliable CI feedback so email bugs are caught before they reach production.            | Technical debt<br>Reliability |
| 5  | #27  | P1       | W1   | Confirm the notification email override reaches the right address end-to-end                             | Guarantees the override setting actually redirects emails -- not just stores the address.       | Reliability                   |
| 6  | #65  | P0       | W1   | Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard         | Removes a technical barrier that currently requires developer access to configure email.        | Client UX<br>Automation       |
| 7  | #66  | P0       | W1   | Confirm a real email reaches the right inbox once Resend is configured for the first client              | End-to-end proof that the Resend integration actually delivers before clients go live.          | Reliability<br>Client UX      |
| 8  | #141 | P1       | W1   | Confirm the welcome email fires when a new user is created -- and only then                              | Prevents duplicate welcome emails from confusing new users during re-invites or re-seeding.     | Reliability                   |
| 9  | #142 | P2       | W3   | Let admins re-send the welcome email to a pending user who hasn't signed in yet                          | Admins can help users who lost their welcome email without creating a new account.              | Client UX<br>Automation       |
| 10 | #143 | P1       | W1   | Make sure welcome emails link to the correct dashboard URL after deployment                              | New users land on the right domain -- not a broken or dev preview link.                         | Reliability                   |

| #  | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                         | WHY IT MAKES THE DASHBOARD BETTER                                                         | TAGS                   |
|----|------|----------|------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------|
| 11 | #202 | P0       | W1   | Alert admins the moment a paid order confirmation email permanently fails to deliver | Paid customers always know their order went through -- failed emails surface immediately. | Revenue<br>Reliability |

8 improvements

## Onboarding & Site Setup

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                 | WHY IT MAKES THE DASHBOARD BETTER                                                               | TAGS                             |
|---|------|----------|------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------|
| 1 | #72  | P1       | W2   | Add an onboarding wizard so admins can seed a new client without touching the CLI            | Non-technical staff can onboard new clients end-to-end through the dashboard UI.                | Automation<br>Client UX          |
| 2 | #73  | P1       | W2   | Confirm the seeder runs end-to-end for a second site without corrupting the first            | Verifies multi-site isolation so onboarding a second client never affects existing data.        | Reliability<br>Security          |
| 3 | #90  | P1       | W2   | Make sure a second launch on the same session can't silently create a duplicate site         | Prevents orphaned duplicate site records from confusing admins and fragmenting data.            | Reliability                      |
| 4 | #91  | P1       | W2   | Confirm the temp domain is assigned to the site record when the client picks a temporary URL | New clients can immediately preview their site at the expected URL after onboarding.            | Client UX<br>Reliability         |
| 5 | #93  | P2       | W2   | Seed placeholder services when onboarding creates a site with services enabled               | New sites launch with example content rather than an empty, confusing services section.         | Client UX                        |
| 6 | #110 | P3       | W4   | Remember that the getting-started nudge was dismissed so it doesn't reappear on every visit  | Eliminates a repetitive UI annoyance that makes the dashboard feel unpolished.                  | Client UX                        |
| 7 | #111 | P3       | W4   | Confirm the placeholder nudge banner shows and disappears at the right times                 | Makes the onboarding experience feel intentional -- guides new users, then gets out of the way. | Client UX                        |
| 8 | #192 | P2       | W2   | Add a CONTRIBUTING guide so new contributors always commit with the right identity           | Prevents messy commit history and identity-check failures from new team members.                | Infrastructure<br>Technical debt |

8 improvements

## Products & Commerce

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                              | WHY IT MAKES THE DASHBOARD BETTER                                                           | TAGS                          |
|---|------|----------|------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------|
| 1 | #107 | P2       | W3   | Let visitors see which product categories exist before filtering                          | Visitors can browse categories without guessing what's available, boosting discoverability. | Client UX Revenue             |
| 2 | #108 | P1       | W2   | Make sure filtering products by category can't silently return wrong results              | Prevents clients from unknowingly showing incorrect product sets to customers.              | Reliability                   |
| 3 | #109 | P1       | W2   | Confirm disabled-module sites can't have their products scraped via the public endpoint   | Protects product data from sites that haven't purchased the commerce module.                | Security                      |
| 4 | #112 | P1       | W2   | Catch a future refactor that breaks the products list endpoint's visibility filter        | Automated check that draft/hidden products can never leak to the public storefront.         | Reliability<br>Technical debt |
| 5 | #114 | P2       | W2   | Confirm seeded product titles always carry the client's business name                     | Demo products feel branded from day one, giving clients a realistic preview of their store. | Client UX                     |
| 6 | #118 | P1       | W2   | Confirm featured products can't appear in the featured spotlight if they're still hidden  | Prevents draft products from accidentally appearing in premium homepage real estate.        | Client UX<br>Reliability      |
| 7 | #123 | P1       | W2   | Prevent two products from sharing the same slug on the same site                          | Eliminates broken product URLs caused by slug collisions that route to the wrong item.      | Reliability                   |
| 8 | #169 | P2       | W3   | Make sure client admins can manage their Square discounts without needing FSTS staff help | Clients can run promotions independently, removing a support dependency on FSTS staff.      | Client UX Revenue             |

3 improvements

# Payments & Checkout

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                        | WHY IT MAKES THE DASHBOARD BETTER                                                                        | TAGS                               |
|---|------|----------|------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------|
| 1 | #203 | P0       | W1   | Confirm Square payments still process correctly end-to-end after the idempotency changes            | Validates that the idempotency hardening didn't break the actual checkout flow for real customers.       | Revenue<br>Reliability             |
| 2 | #204 | P0       | W1   | Harden the provider-agnostic payment webhook with the same idempotency and email-state guards       | Ensures future payment integrations can't double-charge customers or skip order confirmation emails.     | Revenue<br>Reliability<br>Security |
| 3 | #206 | P0       | W1   | Block checkout when a course or event price is missing or corrupt so no broken order reaches Square | Prevents \$0 or malformed orders from reaching Square, protecting clients from disputes and chargebacks. | Revenue<br>Reliability             |

7 improvements

## Courses & Classes

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                     | WHY IT MAKES THE DASHBOARD BETTER                                                                      | TAGS                   |
|---|------|----------|------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------|
| 1 | #157 | P1       | W1   | Confirm courses stay hidden when the agency switches the module off platform-wide                | Module toggles work at the platform level -- clients can't circumvent a module being disabled.         | Security Reliability   |
| 2 | #158 | P1       | W2   | Confirm a disabled-module site can't be unlocked by a direct courses.get call                    | API-level guard prevents clever URL manipulation from bypassing module restrictions.                   | Security               |
| 3 | #189 | P1       | W1   | Confirm lifecycle alerts in the edit dialog always reflect live registration data                | Admins editing a class see accurate enrollment counts, not stale cached data.                          | Client UX Reliability  |
| 4 | #193 | P0       | W1   | Make sure a client with a stale session can't register for a class that's already been cancelled | Prevents ghost registrations where customers pay for a class that no longer exists.                    | Revenue Reliability    |
| 5 | #194 | P1       | W1   | Catch a flyer that stays published after its linked class is deleted outright                    | Orphaned flyers advertising deleted classes are automatically unpublished to avoid customer confusion. | Client UX Reliability  |
| 6 | #195 | P1       | W1   | Confirm the lifecycle clock is registered so tick can never be silently skipped                  | Class status transitions (open -> full -> closed) happen on schedule rather than getting stuck.        | Reliability Automation |
| 7 | #207 | P2       | W3   | Let admins fix a broken course link on an event from the warning banner -- not just see it       | Turns a passive warning about broken course links into a one-click repair action.                      | Client UX Automation   |

3 improvements

Services

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                       | WHY IT MAKES THE DASHBOARD BETTER                                                                     | TAGS                     |
|---|------|----------|------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------|
| 1 | #92  | P0       | W1   | Confirm services created in the dashboard actually appear on the public website    | Closes the gap where a service looks correct in the dashboard but never surfaces on the live site.    | Client UX<br>Reliability |
| 2 | #94  | P1       | W2   | Prevent two services from sharing the same slug on the same site                   | Avoids URL collisions that cause the wrong service page to load for customers.                        | Reliability              |
| 3 | #137 | P1       | W1   | Seed services data for Corsair so the live services section appears on the website | Corsair's website displays a real services section immediately after seeding, no manual entry needed. | Client UX                |

2 improvements

## Flyers & Marketing

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                         | WHY IT MAKES THE DASHBOARD BETTER                                                                               | TAGS                    |
|---|------|----------|------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------|
| 1 | #167 | P2       | W3   | Let clients duplicate an existing flyer instead of rebuilding from scratch           | Dramatically speeds up flyer creation for recurring events -- clients copy and adjust instead of starting over. | Client UX<br>Automation |
| 2 | #168 | P2       | W3   | Confirm health notifications give a clear 'access denied' message to non-admin users | Non-admin users get a clear boundary message instead of an unexplained error.                                   | Client UX<br>Security   |

1 improvements

# Sidebar & Navigation

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                  | WHY IT MAKES THE DASHBOARD BETTER                                                                  | TAGS      |
|---|------|----------|------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------|
| 1 | #176 | P3       | W4   | Add a Role Permissions link to the per-site sidebar so admins can reach it from inside a site | Eliminates navigation friction -- admins can jump directly to role settings from any site context. | Client UX |

4 improvements

## Performance & Build Quality

| # | ID  | PRIORITY | WAVE | WHAT IT DOES                                                                           | WHY IT MAKES THE DASHBOARD BETTER                                                              | TAGS                             |
|---|-----|----------|------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------|
| 1 | #54 | P3       | W4   | Fix the circular vendor chunk warning so every production build is clean               | Clean builds with no warnings make it easier to spot real problems in CI output.               | Technical debt<br>Infrastructure |
| 2 | #55 | P3       | W4   | Make page transitions feel instant by preloading chunks on hover                       | Dashboard navigation feels snappier -- chunks are ready before the user clicks.                | Client UX                        |
| 3 | #83 | P2       | W4   | Extend loading-skeleton guards to the remaining dashboard pages not yet covered        | Eliminates blank-flash loading states on pages that currently show nothing while data loads.   | Client UX                        |
| 4 | #84 | P2       | W2   | Confirm a dropped skeleton guard shows up as a failing test -- not a silent blank page | Prevents future regressions where a missing skeleton silently degrades the loading experience. | Technical debt<br>Reliability    |

2 improvements

CMS & Live Data

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                   | WHY IT MAKES THE DASHBOARD BETTER                                                            | TAGS                     |
|---|------|----------|------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------|
| 1 | #135 | P0       | W1   | Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds | Clients can trust the dashboard price is what customers see -- no manual cache purge needed. | Client UX<br>Reliability |
| 2 | #136 | P1       | W1   | Extend live CMS data to the course detail pages so per-course pricing is also accurate         | Detail pages reflect real pricing, preventing customers from seeing stale rates.             | Client UX<br>Reliability |

3 improvements

# Security & Seeding Safeguards

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                                           | WHY IT MAKES THE DASHBOARD BETTER                                                                     | TAGS     |
|---|------|----------|------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------|
| 1 | #129 | P1       | W2   | Confirm the seed gate can't be bypassed if SEED_ALLOWED is set to 'false' or empty                     | Prevents accidental data seeding on production environments from a misconfigured env var.             | Security |
| 2 | #130 | P1       | W2   | Make sure the add-on catalog seed can't run on production even if someone sets SEED_ALLOWED by mistake | Belt-and-suspenders protection against overwriting live add-on catalog data.                          | Security |
| 3 | #131 | P1       | W2   | Confirm visitors can't forge form submissions for a site they don't belong to                          | Closes a cross-site form injection vector where a malicious visitor could pollute another site's CRM. | Security |

## 6 improvements

## Infrastructure &amp; DevOps

| # | ID   | PRIORITY | WAVE | WHAT IT DOES                                                                         | WHY IT MAKES THE DASHBOARD BETTER                                                                    | TAGS                             |
|---|------|----------|------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------|
| 1 | #20  | P1       | W1   | Make sure Vercel deploys cleanly against the rewritten commit history                | Unblocks deployments that were failing because of the history rewrite required for identity cleanup. | Infrastructure                   |
| 2 | #30  | P2       | W2   | Make the client-website E2E suite actually run in CI against a deployed site         | Catches client-facing website regressions in CI instead of after a client reports them.              | Infrastructure<br>Reliability    |
| 3 | #58  | P2       | W2   | Make sure the workspace branch can't silently drift ahead of origin/main again       | Prevents the confusing scenario where local and remote history diverge without warning.              | Infrastructure<br>Technical debt |
| 4 | #61  | P1       | W1   | Confirm www.fstsclientsystem.com actually redirects once the Vercel alias is added   | Clients can reach the dashboard at the canonical www address -- not just the bare domain.            | Infrastructure<br>Client UX      |
| 5 | #62  | P1       | W2   | Turn on the branch protection rule so the identity check can't be bypassed on GitHub | Enforces commit identity standards at the repository level -- no accidental bypass via direct push.  | Infrastructure<br>Security       |
| 6 | #211 | P0       | W1   | Confirm clients can actually sign in once the Cloudflare DNS fix is applied          | Verifies the DNS fix resolves the sign-in issue for clients on all networks.                         | Client UX<br>Infrastructure      |

## ARCHITECTURAL FOUNDATION

# FSTS-WOS(TM) Multi-Site & E-Commerce Architecture Requirements

## ?? Core Rule

Do not design the FSTS-WOS(TM) around Corsair specifically. Every feature, model, permission, automation, CMS module, API, and component must be evaluated for reuse across many independent client websites.

## ? Client Design Lock

Clients manage approved content, media, products, services, classes, forms, and SEO. They cannot modify source code, structural layouts, design system, backend, or deployment infrastructure. FSTS controls all templates.

## ? Multi-Tenant Isolation

Every record must belong to the appropriate tenant. A client must never read, modify, search, export, or indirectly discover another client's products, orders, customers, payments, registrations, media, leads, or configuration.

## ? Provider-Agnostic Commerce

Square is the first supported payment provider, but the architecture must allow additional providers later. Use provider abstraction for payments, refunds, customers, orders, discounts, and webhooks. Store provider IDs separately from FSTS IDs.

## ? Success Test

**Can we onboard a completely different business -- including a full e-commerce business -- without changing the core dashboard code?** The answer must eventually be YES. Corsair proves #1. Website #2 proves repeatability. An e-commerce client proves platform flexibility. #10 proves scalability.

## ? Checklist for Every Task

1. Is this reusable across tenants?
2. Does it assume Corsair-specific behavior?
3. Does it work for non-Corsair businesses?
4. Does it work when the relevant module is disabled?
5. Does it maintain tenant isolation?
6. Does it respect the client design lock?
7. Could it work for an e-commerce site?
8. Does it scale to dozens or hundreds of sites?

## Supported Business Types (Platform Must Handle All)

Service Businesses

E-Commerce Stores

Training Companies

Course/Class Businesses

Event Businesses

Membership Organizations

Professional Services

Restaurants &amp; Hospitality

Nonprofits

Contractors

Retail Businesses

Product + Service Hybrids

Digital Products

Physical Products

Appointments/Bookings

Subscriptions/Memberships

ROADMAP SUMMARY

Execution Statistics & Top Priorities

83

Total Improvements

18

P0 Critical

41

P1 Required

18

P2 Important

6

P3 Enhancement

36

Wave 1 Items

32

Wave 2 Items

15

Categories

Top 10 Highest-Value Immediate Actions

| #  | ID   | IMPROVEMENT                                                                                         | PRIORITY | RATIONALE                                            |
|----|------|-----------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|
| 1  | #211 | Confirm clients can actually sign in once the Cloudflare DNS fix is applied                         | P0       | Clients cannot sign in -- blocking all revenue       |
| 2  | #203 | Confirm Square payments still process correctly end-to-end after the idempotency changes            | P0       | Validates payment flow after idempotency changes     |
| 3  | #206 | Block checkout when a course or event price is missing or corrupt so no broken order reaches Square | P0       | Stops \$0/corrupt orders reaching Square             |
| 4  | #204 | Harden the provider-agnostic payment webhook with the same idempotency and email-state guards       | P0       | Protects all future payment integrations             |
| 5  | #92  | Confirm services created in the dashboard actually appear on the public website                     | P0       | Services not appearing on live website (#1 CMS gap)  |
| 6  | #202 | Alert admins the moment a paid order confirmation email permanently fails to deliver                | P0       | Paid customers must receive order confirmation       |
| 7  | #65  | Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard    | P0       | Email config requires developer access today         |
| 8  | #193 | Make sure a client with a stale session can't register for a class that's already been cancelled    | P0       | Ghost registrations = payments for cancelled classes |
| 9  | #85  | Make the guard check smart enough to catch null checks that don't show an access-denied message     | P0       | Blank screens instead of access-denied messages      |
| 10 | #135 | Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds      | P0       | Price changes must reflect within 60 seconds         |